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Supplementary appendix 2

This appendix formed part of the original submission and has been peer reviewed. We post it as supplied by the authors.

Supplement to: Xu R-H, Zhang Y, Pan H, et al. Efficacy and safety of weekly paclitaxel with or without ramucirumab as second-line therapy for the treatment of advanced gastric or gastroesophageal junction adenocarcinoma (RAINBOW-Asia): a randomised, multicentre, double-blind, phase 3 trial. *Lancet Gastroenterol Hepatol* 2021; published online October 6. https://doi.org/10.1016/S2468-1253(21)00313-7.

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Supplementary Methods

Criteria for dose modification/discontinuation of ramucirumab

- Dose modifications were permitted for ramucirumab/placebo in a non-life-threatening setting and if reversible grade 3 clinical adverse events (AEs) (for example, fever) considered to be at least possibly related to ramucirumab/placebo resolved to grade ≤1 or pretreatment baseline within 1 treatment cycle (approximately 28 days).
- If a grade 4 AE occurred and was deemed at least possibly related to ramucirumab/placebo, then ramucirumab/placebo was discontinued except in the specific case of grade 4 fever or grade 4 laboratory abnormalities. If grade 4 fever or laboratory abnormalities resolved to grade ≤1 or pretreatment baseline within 1 treatment cycle, treatment with the investigational product could be continued at the discretion of the investigator. In these settings, ramucirumab/placebo could be readministered.
- If the second instance of such an event occurred, ramucirumab/placebo was subsequently readministered at a dose of 6 mg/kg every other week. A second dose reduction to 5 mg/kg every other week was permitted for this level of event (grade 3 or 4 event). If the dose of ramucirumab/placebo was reduced because of potentially treatment-related AEs, subsequent dose increases were not permitted.
- Ramucirumab/placebo was to be permanently discontinued in case of any ramucirumab/placebo-related event that was deemed life-threatening, regardless of grade.

Criteria for dose modification/discontinuation of paclitaxel

- No dose modification for paclitaxel was allowed within a given cycle. The paclitaxel dose was reduced by 10 mg/m^2 for the following cycle when National Cancer Institute Common Terminology Criteria for Adverse Events version 4.03, grade 4 haematological toxicity or grade 3 paclitaxel-related nonhematological toxicity (except for alopecia) was observed.
- If the dose of paclitaxel was reduced because of potentially related AEs, subsequent dose increases were not permitted.
- Paclitaxel was to be permanently discontinued if dose reduction to less than 60 mg/m^2 was required, or in case of any paclitaxel-related event that was deemed life-threatening, regardless of grade.

Table S1. Post-discontinuation therapies

	Ramucirumab plus paclitaxel N = 294	Placebo plus paclitaxel N = 146
Surgery	15 (5·1%)	7 (4·8%)
Radiotherapy	11 (3·7%)	10 (6·8%)
Systemic therapy	160 (54·4%)	82 (56·2%)
Chemotherapy	95 (32·3%)	50 (34·2%)
Targeted therapy [a]		
Immunotherapy	43 (14·6%)	18 (12·3%)
Anti-angiogenesis	71 (24·1%)	41 (28·1%)
Anti-HER2	17 (5·8%)	4 (2·7%)
Traditional Chinese medicine	24 (8·2%)	7 (4·8%)
Other	7 (2·4%)	5 (3·4%)
Received ≥2 subsequent lines	60 (20·4%)	30 (20·5%)
Received ≥3 subsequent lines	21 (7·1%)	12 (8·2%)

Data are n (%).

[a] Included all lines of post-discontinuation systemic therapy received.

Abbreviation: HER2, human epidermal growth factor receptor 2

Table S2. Sensitivity analysis for progression-free survival

Progression-free survival	Hazard ratio (95% confidence interval)[a,b]
Analysis 1	0·776 (0·621–0·968)
Analysis 2	0·791 (0·636–0·984)
Analysis 3	0·794 (0·639–0·985)
Analysis 4	0·767 (0·615–0·957)
Analysis 5	0·770 (0·617–0·961)
Per protocol analysis	0·771 (0·617–0·964)
Adjusted for prognostic factors using a stepwise Cox regression[c]	0·714 (0·572–0·892)

[a] Stratified by ECOG PS and peritoneal metastases.
[b] Estimated using the Cox model.
[c] The final covariates in the model were: ECOG PS, Peritoneal metastases, liver metastasis, and the number of previous treatment lines. The hazard ratio for treatment effect was estimated using an unstratified multivariate Cox proportional hazard model.
Note: Analysis 1: Included cases of symptomatic deterioration as progression events; Analysis 2: Patients who began a new anticancer therapy prior to progression or death were not censored; Analysis 3: If death or disease progression occurred after ≥2 consecutively missed radiographic assessment visits, the patient was considered to have progressed at the date of the first missed tumour assessment; Analysis 4: Patients lost to follow-up (as recorded in the CRF for End of Study reason) without documented progression or death were considered to have disease progression at the date of the next scheduled visit after the last tumour assessment; Analysis 5: In the ramucirumab plus paclitaxel group, patients lost to follow-up (as recorded in the CRF for End of Study reason) without documented progression or death were considered to have progression at the date of the next scheduled visit after the last tumour assessment.
Abbreviation: ECOG PS, Eastern Cooperative Oncology Group performance status.

Table S3. Sensitivity analysis for overall survival

Overall Survival	Hazard ratio (95% confidence interval)[a,b]
Adjust for prognostic factors using stepwise Cox regression [c]	0·954 (0·762–1·193)
Per-protocol analysis	0·981 (0·785–1·227)
Censoring users of PDT to the start date of PDT [d]	0·831 (0·589–1·172)
Censoring users of ≥2 lines PDT [d]	0·854 (0·668–1·092)
Censoring users of ≥3 lines PDT [d]	0·896 (0·713–1·126)
Censoring users of targeted PDT [d, e]	0·869 (0·658–1·148)
Censoring users of anti-angiogenesis PDT [d]	0·942 (0·725–1·223)

[a] Stratified by ECOG PS and peritoneal metastases.
[b] Estimated using the Cox model.
[c] The final covariates in the model were: ECOG Performance Status, Peritoneal Metastases, Country, PDT use, Liver metastasis, Tumor differentiation at initial diagnosis, and Prior gastrectomy. The hazard ratio for treatment effect was estimated using an unstratified multivariate Cox proportional hazard model.
[d] OS for the corresponding user of PDT were censored at the start date of PDT.
[e] Targeted PDT included immunotherapy, anti-angiogenesis, anti-HER2 drug.
Abbreviation: ECOG PS, Eastern Cooperative Oncology Group performance status; PDT, post-discontinuation systemic therapy.

Table S4. Quality of life QLQ-C30 and EQ-5D-3L mean scores at baseline and short-term follow up

	Ramucirumab plus paclitaxel N = 294	Placebo plus paclitaxel N = 146
EORTC QLQ-C30		
Patients completing baseline QLQ-C30	294 (100%)	146 (100%)
Baseline global quality of life score[a]	68·9 (20·6)	69·5 (18·8)
Patients completing baseline plus post-baseline QLQ-C30	272 (92·5%)	133 (91·1%)
Patients completing QLQ-C30 during short-term follow-up	123 (41·8%)	56 (38·4%)
Global quality of life score during short- term follow-up[a]	59·6 (21·3)	60·1 (23·5)
EQ-5D-3L		
Patients completing baseline EQ-5D-3L	294 (100%)	146 (100%)
Baseline index score[b]	0·86 (0·15)	0·89 (0·13)
Patients completing baseline plus post-baseline EQ-5D-3L	272 (92·5%)	133 (91·1%)
Patients completing EQ-5D-3L during short-term follow-up	123 (41·8%)	56 (38·4%)
Index score during short-term follow-up[b]	0·73 (0·25)	0·78 (0·21)

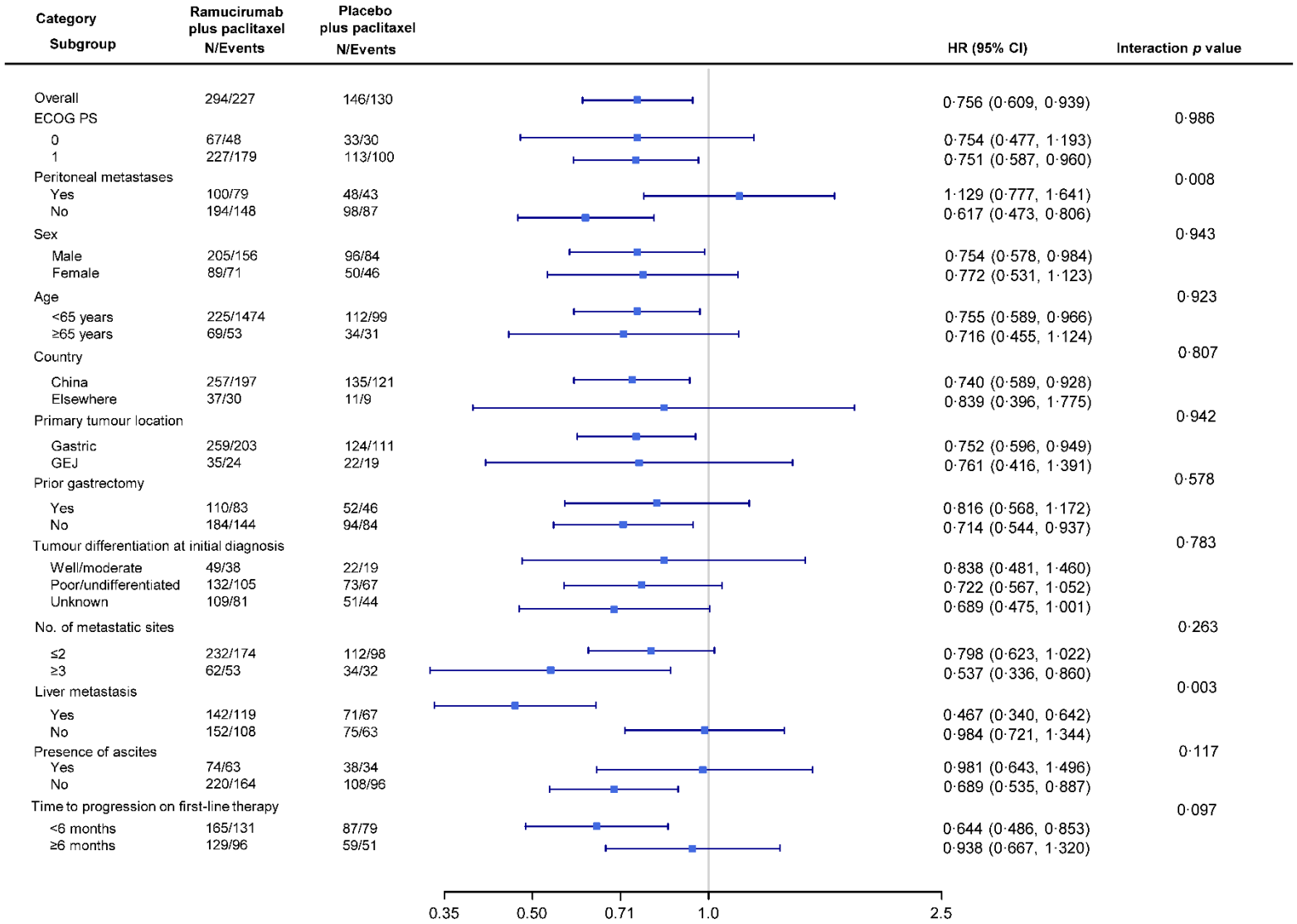
Data are n (%) or mean (standard deviation).

[a] Based on a 100-point scale, with a higher score representing better quality of life.

[b] Based on a − 0·149 to 1 scale, with 1 representing perfect health.

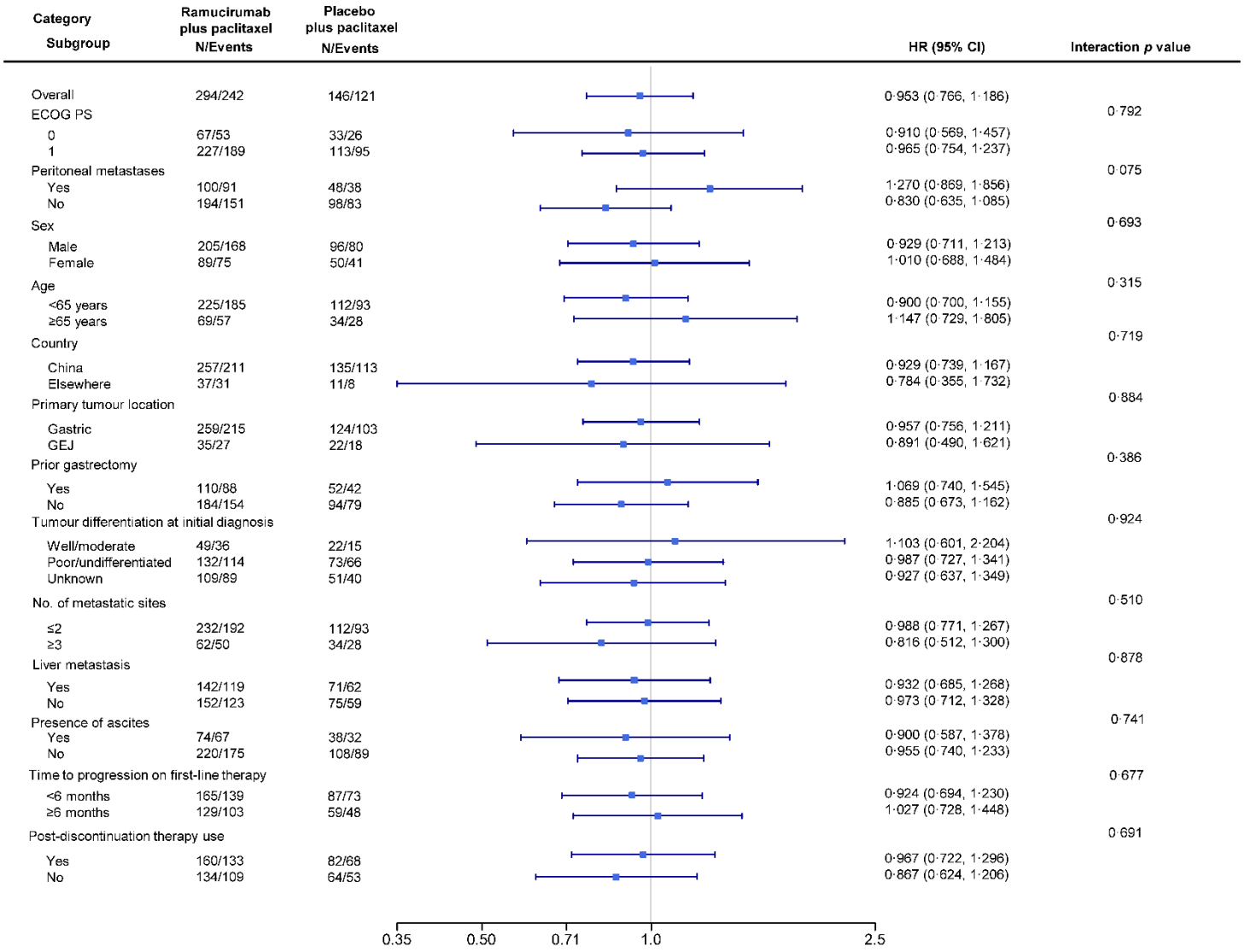
Abbreviation: EORTC QLQ-C30, European Organization for Research and Treatment of Cancer Quality of Life Questionnaire Core 30.; EQ-5D-3L, EuroQol 5-Dimension 3-Level Questionnaire.

Figure S1. Forest plot for subgroup analysis of progression-free survival



Abbreviations: CI, confidence interval; ECOG PS, Eastern Cooperative Oncology Group performance status; GEJ, gastroesophageal junction; HR, hazard ratio.

Figure S2. Forest plot for subgroup analysis of overall survival



Abbreviations: CI, confidence interval; ECOG PS, Eastern Cooperative Oncology Group performance status; GEJ, gastroesophageal junction; HR, hazard ratio.

List of Investigators

Country	Site Reference Number	Principle Investigator Name	City/Province	Primary Centre	No. of Patients Screened	No. of Patients Randomized
China	110	Lin Shen	Beijing	Beijing Cancer Hospital	92	67
China	130	Yanqiao Zhang	Harbin, Heilongjiang	Harbin Medical University Cancer Hospital	60	54
China	111	Ruihua Xu	Guangzhou, Guangdong	Sun Yat-sen University Cancer Center	50	38
China	120	Hongming Pan	Hangzhou, Zhejiang	Sir Run Run Shaw Hospital, School of Medicine, Zhejiang University	35	31
China	115	Tao Zhang	Wuhan, Hubei	Union Hospital of Tongji Medical College Huazhong University of Science and Technology	30	23
China	116	Jifeng Feng	Nanjing, Jiangsu	Jiangsu Cancer Hospital	27	24
China	113	Tianshu Liu	Shanghai	Zhongshan Hospital Fudan University	26	22
China	114	Yanru Qin	Zhengzhou, Henan	The First Affiliated Hospital of Zhengzhou University	24	20
China	123	Yi Ba	Tianjin	Tianjin Medical University Cancer Institute & Hospital	22	12
China	128	Shukui Qin	Nanjing, Jiangsu	Cancer Center of Bayi Hospital, Nanjing Chinese Medicine University (The No.81 Hospital of PLA)	20	16
China	112	Baorui Liu	Nanjing, Jiangsu	Nanjing Drum Tower Hosp Affiliated Hosp of Nanjing Univ Med	19	14
China	132	Xianli Yin	Changsha, Hunan	Hunan Cancer Hospital	19	15
China	119	Helong Zhang	Xi'an, Shanxi	The Chinese PLA, Fourth Military Medical University, Tangdu Hospital	14	8
Tailand	140	Suebpong Tanasanvimon	Pathumwan, Bangkok	King Chulalongkorn Memorial Hospital	12	8
China	121	Nong Yang	Changsha, Hunan	Hunan Cancer Hospital	11	10
China	133	Ying Cheng	Changchun, Jilin	Jilin Cancer Hospital	11	9
Malaysia	156	Ibtisam Muhamad Nor	Kuala Lumpur	Hospital Kuala Lumpur	11	7
Malaysia	160	Kong Leong Yu (2016.11-2018.10) Pei Jye Voon (2018.10-2020.08) Swee Kiong Kho (2020.08-current)	Kuching	Hospital Umum Sarawak	10	8
China	129	Xianglin Yuan	Wuhan, Hubei	Tongji Hospital of Tongji Medical College Huazhong University of Science and Technology	9	7
China	117	Xiaoyan Lin	Fuzhou, Fujian	Fujian Medical University Union Hospital	8	5
China	126	Yunpeng Liu	Shenyang, Liaoning	The First Hospital of China Medical University	8	5
Philippines	151	Adonis Guancia	Bacolod	Dr. Pablo O. Torre Memorial Hospital	7	5
Tailand	142	Jedzada Maneechavakajorn	Ratchathewi, Bangkok	Rajavithi Hospital	6	4

Malaysia	158	Florence Yoke Hui Wong	Putrajaya	National Cancer Institute	6	5
China	118	Ying Yuan	Hangzhou, Zhejiang	The Second Affiliated Hospital of Zhejiang University School of Medicine	5	3
China	122	Dong Ma	Guangzhou, Guangdong	Guangdong Provincial People's Hospital	5	4
China	125	Jianming Xu	Beijing	The Fifth Medical Center of PLA General Hospital (The 307th Hospital of Chinese PLA)	5	5
Malaysia	159	Gwo Fuang Ho	Kuala Lumpur	University Malaya Medical Centre	5	5
Tailand	143	Sureerat Jaruhathai	Pathumwan, Bangkok	Police General Hospital	2	2
Philippines	150	Ma. Noemi Uy	Cebu	Cebu Doctors Hospital	2	2
Tailand	141	Luangyot Thongthieang	Tambon Nai Mueang, Amphoe Mueang Khon Kaen	Khon Kaen Hospital	1	1
Philippines	148	Paul Dexter Santos	Dasmariñas, Cavite	De La Salle Health Sciences Institute	1	1